LOGISTICSOLVER 2026 · RESEARCH PROTOTYPE

# Fee cliffs.
# Better schedules.
# Priced resilience.

PortPlan — container pickup, unloading and empty-return planning under public Korean tariff rules.

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Focused question: when do exact fee rules change a decision, and what does a buffer cost?

# One extra day can be expensive.

**₩318,500**

PNC storage · 20 ft · day 14

**₩500,500**

PNC storage · 20 ft · day 15

**+₩182,000**

The extra day crosses a retroactive tier

PNC applies the highest reached rate to **all chargeable days beyond the four free days**. A smooth daily-cost approximation can miss this jump.

₩318,500 = 10 × 31,850; ₩500,500 = 11 × 45,500. Separate Maersk D&D continues until empty return.

Source: PNC Standard Tariff 2026, p.4; Maersk PNC storage notice. Standard dry only, taxes/contracts excluded.

# Plan the full container lifecycle.

**Release** → **Pickup** → **Unload** → **Empty return**

## Decisions and constraints

- One pickup, unloading start and return date per box
- Daily movement, unloading and retained-box capacity
- Release dates and hard unloading deadlines

## What the model excludes

- Road routes, driver hours and appointment slots
- Negotiated contracts, taxes and transport prices
- Online information and already-executed actions

Objective: terminal storage + carrier D&D. “Optimal” means optimal within this stated static daily model.

# Separate rules, assumptions and observations.

| Evidence | What it supports | What it cannot prove |
|---|---|---|
| PNC 2026 official tariff | Retroactive storage calculation | Every shipment's negotiated bill |
| Maersk Korea official example | 10 inclusive free days; first-band charge | Cross-band accumulation is still an interpretation |
| 58,171 public Gwangyang gate records | Observed gate-duration context | PNC orders, return chains or fee savings |
| Generated order scenarios | Controlled, reproducible comparisons | Realized enterprise savings |

Older PNC inquiry-page prices conflict with the dated 2026 PDF. The source conflict remains visible.

Sources and applicability: source_registry.json; source-audit.md; rules-status-v2.md.

# An auditable local application. No algorithmic first claim.

## Prior work already exists

- Jeong et al. (2020): devanning and detention
- Jeong et al. (2026): uncertain D&D decisions
- Drayage MILP, release/due dates and empty-stock coordination
- PortPro: pickup/return appointment matching

## Contribution to test

- Encode PNC's exact 2026 fee cliff
- Audit the complete pickup/return cost chain
- Expose sensitivity, failures and buffer premiums
- Publish data, code and independent checks

Nine research works and three products reviewed at disclosed depths. A focused search does not establish global novelty.

References: prior-art-v2.md. DOI 10.1016/j.tre.2020.102055; 10.1111/itor.13622.

# Freeze the questions before the run.

**48**

New seed scenarios · 16 / 50 boxes

**3**

Backlog / new arrivals / batch release

**6**

Policies under identical inputs and constraints

**Stronger internal comparator:** earliest-deadline rule, 12 randomized permutations and 80 relocate/swap attempts, all with a complete greedy decoder.

**Transparent budgets:** fixed search attempts versus a two-second MILP limit. Wall time is reported; budgets are not claimed equal.

Four 100-box controls and 56 sensitivity/ablation runs are separate. All cases retained.

Main protocol frozen in commit 1fb4678 before runs. Enhanced internal baseline, not a reproduced state-of-the-art algorithm.

# Value is concentrated, not universal.

**44/48**

MILP complete feasible · 33 optimal / 11 incumbent

**36/48**

Stronger search complete feasible

**15.74%**

Lower incremental two-fee sum · 36 paired cases

Total two-fee reduction for those same pairs: **8.86%**. Nineteen improve; seventeen tie. Twelve cases lack a complete baseline and are not assigned invented savings.

> All 14 paired new-arrival cases cost zero under both methods. The value case is fee-exposed backlog and concentrated release.

Main experiment: synthetic inputs; four model-infeasible cases retained. Cost-weighted descriptive aggregate, not an expected business saving.

# Check why a plan changes.

| Deliberate model change | Extra cost under the same reference tariff | Interpretation limit |
|---|---|---|
| Smooth PNC as progressive tiers | ₩185,200 / 2.22% across 6 pairs | Evidence from a small fixed panel |
| Optimize storage only | ₩87,550 across 5 strict pairs | Exclude one unresolved secondary tie-break |
| Optimize D&D only | ₩487,550 across 6 pairs | Entire difference comes from one case |

Single-objective runs first minimize their own fee, then use remaining budget to minimize joint fees among ties. Unfinished tie handling stays visible.

Custom free days shift tariff tiers in this research model. That is not a verified contract-extension rule.

Sensitivity is paired scenario evidence, not proof of a universal mechanism or actual contractual exposure.

# Keep the feasible plan when MILP times out.

## 100 boxes

**New arrivals · tight capacity**

Strong search found a complete zero-fee plan. MILP returned no incumbent within two seconds.

**Decision-support correction**

PortPlan recommends the lowest-cost complete audited candidate among the available policies.

It preserves each raw solver status and does not label a fallback globally optimal.

> A time limit is not an infeasibility proof. No incomplete plan enters the fee comparison.

Counterexample retained in scaleControls. This is a portfolio selection rule, not a new optimality guarantee.

# A static optimum can be fragile.

**16/18**

Original plans invalid under fixed disruptions

**18/18**

Hindsight re-solves feasible

**≠ online**

Future disruptions were known in re-solves

Fixed tests reduced daily transport by two on days 3–5, extended every fourth box's unloading by one day, or combined both changes.

Six base scenarios are reused across disruptions. These are dependent stress cases, not a sampled field failure rate.

Independent audit uses actual processing durations and capacities, not cached completion fields.

# Resilience has a measurable premium.

**16 → 0**

Invalid plans across the 18 specified checks

**+32.29%**

Total two-fee premium across 6 pairs

**₩4,652,550**

Additional modeled fees

**Buffer rule:** add one unloading day to every box and reserve two transport movements on every planning day. It does not select the disrupted boxes or dates.

Incremental-fee premium: 76.88% — a different denominator. Three new-arrival cases add no fees; backlog cases bear the premium.

> Exploratory study designed after observing failures. Valid only inside the stated duration/capacity envelope and daily model.

# Make the decision inspectable.

## Three-minute demonstration

1. Verify the day-14 → day-15 fee cliff.
2. Load a fixed experiment; compare six complete schedules.
3. Inspect a failure, then the buffer-cost tradeoff.
4. Export the scenario, assumptions and plan.

## Before a final submission

- Confirm solo entry, English presentation and AI conditions
- Validate shipment-level inputs and contract applicability
- Reproduce a suitable external benchmark
- Explain every assumption and source in Q&A

The claim: transparent fee-aware decisions and bounded resilience tradeoffs — with the evidence and failure cases exposed.